HPAStainR: an R package and Shiny app to query protein patterns in the Human Protein Atlas

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Introduction

The Human Protein Atlas[1] (HPA) is a publically available resource that has immunohistochemistry-based visual proteomics data for over 15,313 proteins across 59 tissues. Within these tissues pathologists have scored what cell types are marked by these stains. This is a large amount of data that can be used to classify gene or protein lists into different cell types, resulting in a powerful resource allowing one to use gene and protein lists to find cell type enrichment. HPA has their own website, but as it stands, it only allows individuals to query one protein at a time, limiting the rate one can explore the data with.

We introduce HPAStainR, a Bioconductor R package and Shiny app developed to query the cell staining data of the HPA. HPAStainR allows user to input a list of proteins or genes and returns a rank ordered list of cell types that are stained for the input list, with various customizations.

Workflow in R

Acquire HPA data through hpa_data_downloader() or by the data function after loading the hpar library

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Provide the HPA tissue data and a gene list to HPAStainR() or shiny_HPAStainR() for a local instance of the Shiny App

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Data frame of cell types ordered on how many proteins stained said cell types and what intensity

Options

- Normal tissue and/or cancer tissue
- Select result confidence levels
- Fisher Test or Chi Square Test
- Column name stylization
- etc.

Use Cases and Examples

A

HPAStainR Output | Summary of Proteins for Cell Types

Show 25 entries | Search:

Cell Type	Percent High Expression	Percent Medium Expression	Percent Low Expression	Percent Not Detected	Number of Proteins	Staining Score	Tested Proteins	Detected Proteins	P-Value	P-Value Adjusted
LUNG - pneumocytes	17.0%	13.0%	17.0%	52.0%	23	27.75	ALOX15B, C4BPA, CPB2, CRTAC1, DMBT1, FGA, FGFBP1, FGG, FUT3, HKDC1, LAMP3, LRP2, PCSK2, PGC, PIGR, PLA2G3, PLA2G4F, SFTPA1, SFTPA2, SFTPC, SLC26A9, SLC6A14, SLC6A20	ALOX15B, CRTAC1, DMBT1, LAMP3, PIGR, PLA2G3, SFTPA1, SFTPA2, SFTPC, SLC6A14, SLC6A20	7.207277e-08	5.909967e-06

B

HPAStainR Output | Summary of Proteins for Cell Types

Show 25 entries | Search:

Cell Type	Percent High Expression	Percent Medium Expression	Percent Low Expression	Percent Not Detected	Number of Proteins	Staining Score	Tested Proteins	Detected Proteins	P-Value	P-Value Adjusted
BRONCHUS - respiratory epithelial cells	56.0%	32.0%	6.0%	6.0%	34	73.50	ADGB, ALOX15, ARMC3, BPIFA1, BPIFB1, CAPSL, CCDC114, CDHR4, CHST9, DNAAF1, DNAH12, DNAH9, DNAI1, DNAI2, DYDC2, EFCAB1, ERN2, FAM183A, FAM92B, GSTA1, MS4A8, MUC16, MUC5B, RSPH1, SCGB1A1, SNTN, SPAG6, STMND1, STOML3, TEKT1, TEKT2, TEKT4, VWA3B, ZBBX	ADGB, ALOX15, ARMC3, BPIFB1, CAPSL, CCDC114, CDHR4, CHST9, DNAAF1, DNAH12, DNAH9, DNAI1, DNAI2, DYDC2, EFCAB1, FAM183A, FAM92B, GSTA1, MS4A8, MUC16, MUC5B, RSPH1, SCGB1A1, SNTN, SPAG6, STMND1, STOML3, TEKT1, TEKT2, TEKT4, VWA3B, ZBBX	2.141124e-23	1.584432e-21

Figure 1. Recapitulated results of gene list previously associated with type II pneumocytes and bronchial epithelial cells respectively[2].

Use Case 1. During bulk –Omics analysis gene and protein lists are often generated. Current GO and meta-analysis tools do not use the deep immunohistochemistry data found in the Human Protein Atlas which can help clarify if a gene/protein list is associated with a certain cell type. For example Fig. 1 represents the findings of inputting two gene lists from known high variance gene clusters, which HPAStainR correctly characterizes as a pneumocyte signal and a bronchus epithelial signal.

Equations

$$\text{Staining Score} = \left(\frac{h \times 100}{t}\right) + \left(\frac{m \times 50}{t}\right) + \left(\frac{l \times 25}{t}\right)$$

Equation 1. Where *t* is the total number of proteins from the list tested in the cell type, *h* is the number of proteins with high staining in the cell type, *m* is the number of proteins with medium staining in the cell type, and *l* is the number of proteins with low staining in the cell type.

Table 1. A subset of 10 HPAStainR results of PanglaoDB cell type marker queries.

PanglaoDB Cell Type	Confidence Score	Top HPAStainR Result	Top Result Stained Score	Top Adjusted P-value	Select Tissue Top Result	Select Tissue Top Stained Score	Select Tissue Adjusted P-value
KIDNEY – proximal tubule cells	66.75	KIDNEY – cells in tubules	66.75	1.6E-18	KIDNEY - cells in tubules	66.75	1.59E-18
HEART MUSCLE – Cardiomyocytes	61	HEART MUSCLE – myocytes	61	1.3E-33	HEART MUSCLE - myocytes	61	1.29E-33
IMMUNE SYSTEM – Neutrophils	60	BONE MARROW – hematopoietic cells	60	1.0E-12	BONE MARROW - hematopoietic cells	60	9.99E-13
OLFACTORY SYSTEM – olfactory epithelial cells	51.25	BRONCHUS – respiratory epithelial cells	51.25	2.2E-05	NASOPHARYNX - respiratory epithelial cells	39.75	0.00076
CONNECTIVE TISSUE – adipocytes	33.75	KIDNEY – cells in tubules	33.75	1	ADIPOSE TISSUE - adipocytes	24.25	0.02353
CONNECTIVE TISSUE – chondrocytes	27.25	TONSIL – squamous epithelial cells	27.25	1	SOFT TISSUE - chondrocytes	20.75	0.00025
REPRODUCTIVE – granulosa cells	13.6	PLACENTA – trophoblastic cells	42.5	0.00174	OVARY - follicle cells	25	1
HEART MUSCLE – purkinje fiber cells	5.775	CAUDATE – neuronal cells	57.75	1	HEART MUSCLE - myocytes	0	1
BRAIN – cholinergic neurons	5.25	DUODENUM – glandular cells	37.5	1	CEREBRAL CORTEX - neuronal cells	31.25	1
EMBRYO – trophoblast progenitor cells	3	PLACENTA – trophoblastic cells	50	1	tissue not found	NA	NA

Use Case 2. Similarly to bulk sequencing, single cell sequencing often results in lists of marker genes. In Table 1 we look at how well HPAStainR indicates the correct cell type for a list of curated marker genes from PanglaoDB[3]. We selected cell types that HPAStainR performs well on, but also those it performs poorly on. The confidence score is a scaled version of the staining score, accounting for how many proteins are in the list, the more proteins the more confident we are in the results. Cell types with larger marker lists appear to perform well. Medium sized lists can call the wrong cell types by staining score, but the correct cell type is called by the fisher test. There are however cell types, due to small lists or lack of HPA data, that simply do not perform well.

Paper QR Code

Shiny App QR Code

References

1. The Human Protein Atlas : https://www.proteinatlas.org/humanproteome/tissue+specific
2. Analysis of the GTEx Lung Transcriptome: PMID: 27588449
3. PanglaoDB: https://panglaodb.se/v